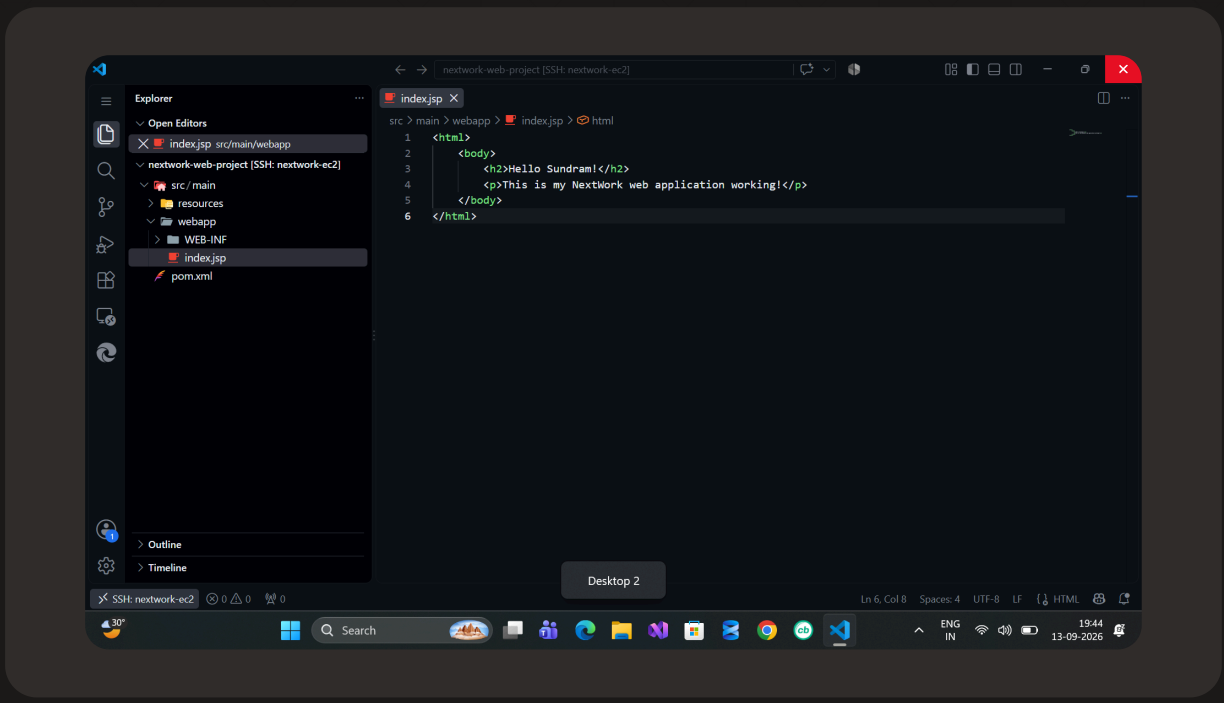
SET UP A WEB APP IN THE CLOUD

By Sundram Singhal | September 2026



Introducing Today's Project!

In this project, I will demonstrate... I'm doing this project to learn basics of building a AWS app along with Vs Code

This project is part one of a series of DevOps projects where I'm building a CI/CD pipeline! I'll be working on the next project from tomorrow itself.

I did this project because i wanna learn DEVOPS. and wanna add one more skill in my reusme.

Key tools and concepts

Services I used were... Key concepts I learnt include creating my ec2 instance and then connectinf that instance with my vs code and vs code terminal.

Project reflection

This project took me approximately... The most challenging part was connceting my instance with the vs code as it was disconnecting again and again. it cost me 4 hours but later on i solved the problem.

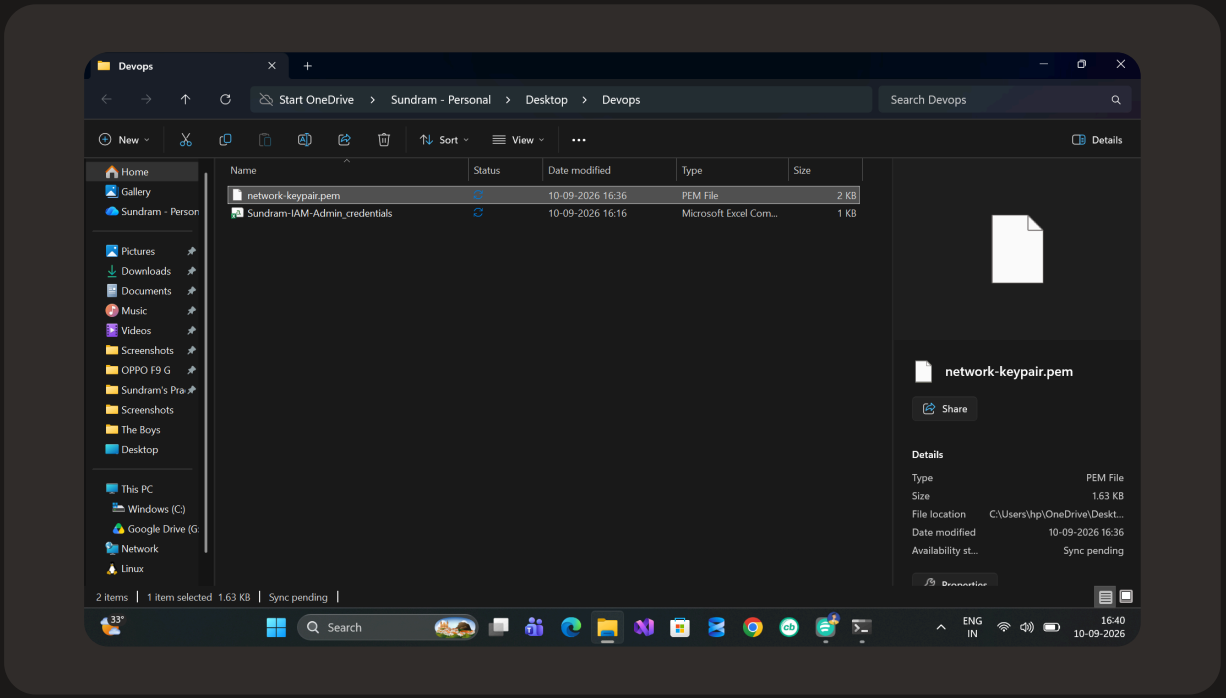
One thing I didn't expect in this project was connecting with vs code also it cost me 4 hours to connect as there was RAM problem which i solved by reading documentation itself.

Launching an EC2 instance

What I did in this step

In this step, I will Launch a new EC2 instance, Set up a key pair for secure access and set up network settings for my instance.

I started this project by launching an EC2 instance because i want to use AWS service which will allow me to rent a server instead of aquiring it on my local computer physically



I also enabled SSH

SSH is... I enabled SSH so that only authorized user can access a remote server. When i will connect to my server later it will verify that i have the correct private key that matches the public key on the server.

Key pairs

A key pair in EC2 is like the keys to your virtual computer, It lets you securely access your EC2 instance.

Downloaded key pair file

Once I set up my key pair, AWS automatically downloaded .pem file which contains my private key

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Set up VS Code

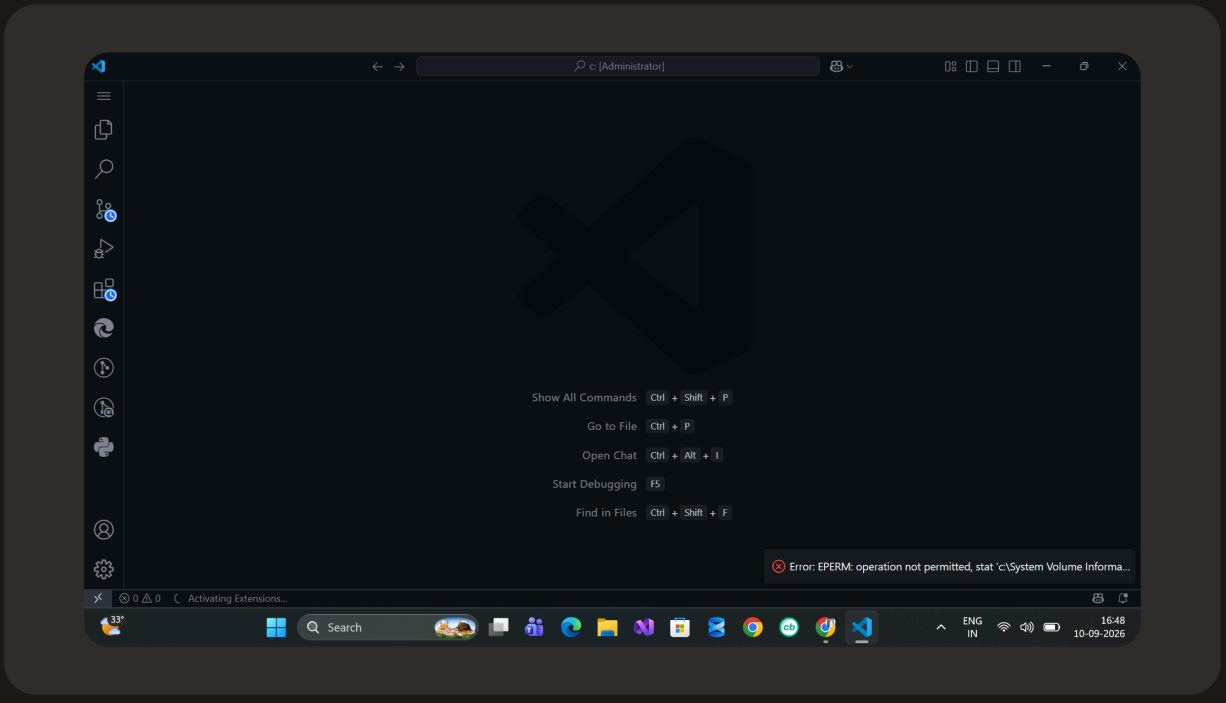
What I did in this step

In this step, I will installl vs code because i want to connect with the instance i have created.

What is VS Code?

VS Code is an IDE where i will connect with my created instance.

I installed VS Code to connect with my instance and create my first web app.

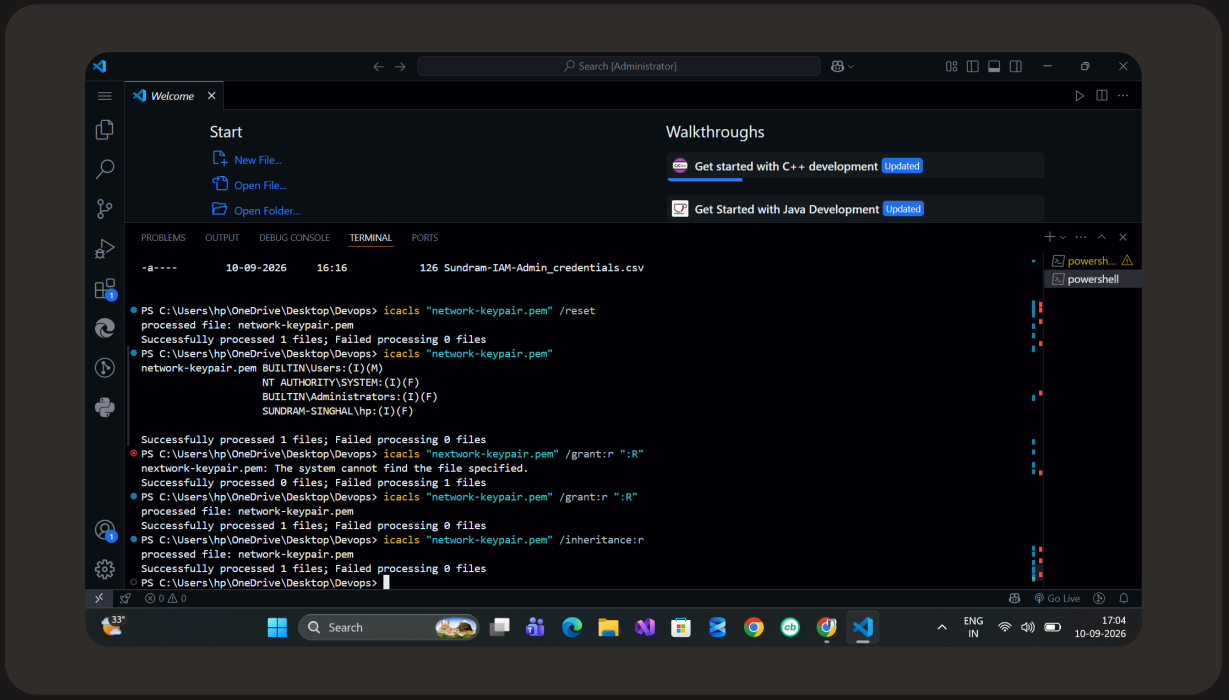


My first terminal commands

A terminal is a way to talk to our computer by instructions instead of clicks. The first command I ran for this project is cd OneDrive

Updating file permissions

I also updated my private key's permissions by using 3 commands and to ensure only i can access that.



SSH connection to EC2 instance

What I did in this step

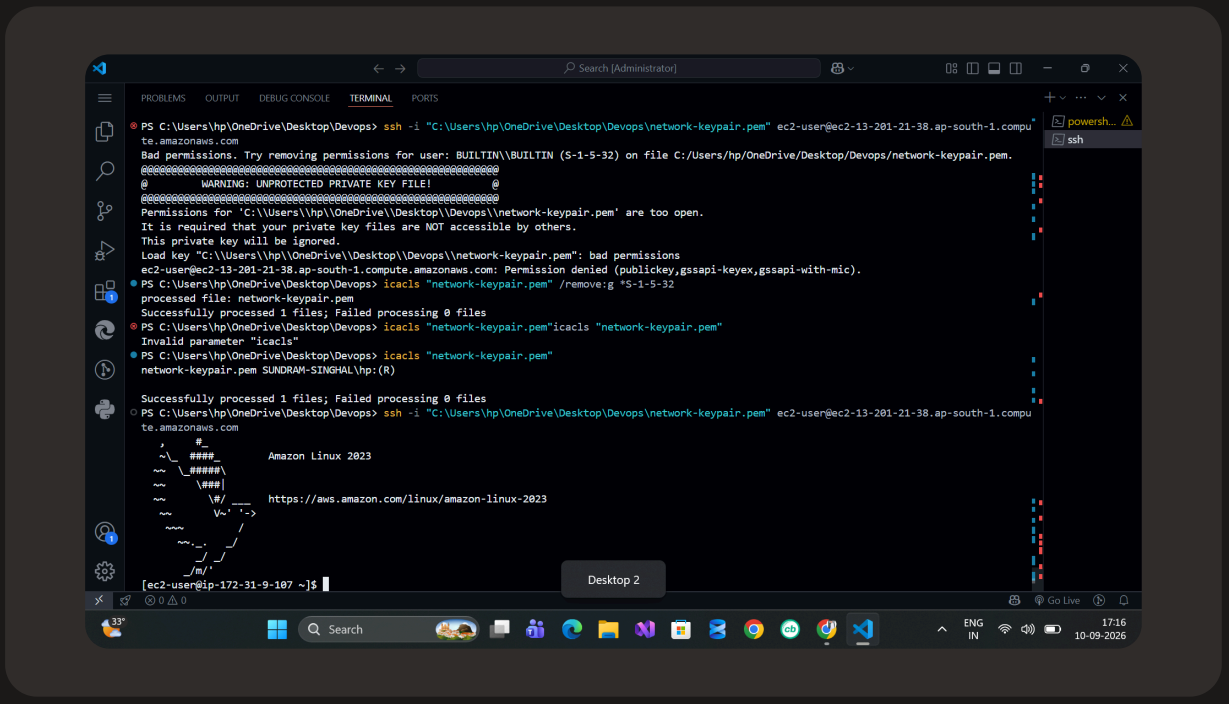
In this step, I will connect to our EC2 instance because i want to work inside my EC2 instance to set up that web app.

Connecting to EC2

To connect to my EC2 instance, I ran the command listed above

This command required an IPv4 address

A server's IPV4 DNS is the public address for my EC2 server that the internet uses to find and connect to it.



Maven & Java

What I did in this step

In this step, I will install apache maven on our EC2 instance, install amazon corretto 8, a version of java and verify the installations.

Why I'm using Maven

Apache Maven is a tool that helps developers build and organize Java software projects. It's also a package manager, which means it automatically download any external pieces of code your project depends on to work.

Maven is required in this project because the application is a Java application, and Maven helps us build, manage dependencies, and package the application.

Why I'm using Java

Java is a programming langauge that allows us to build different applications. Maven required java to build

Java is required in this project because maven required it to operate.

Create the Application

What I did in this step

In this step, I will run maven commands in my terminal because i want to generate a Java web app.

Creating the Java web app

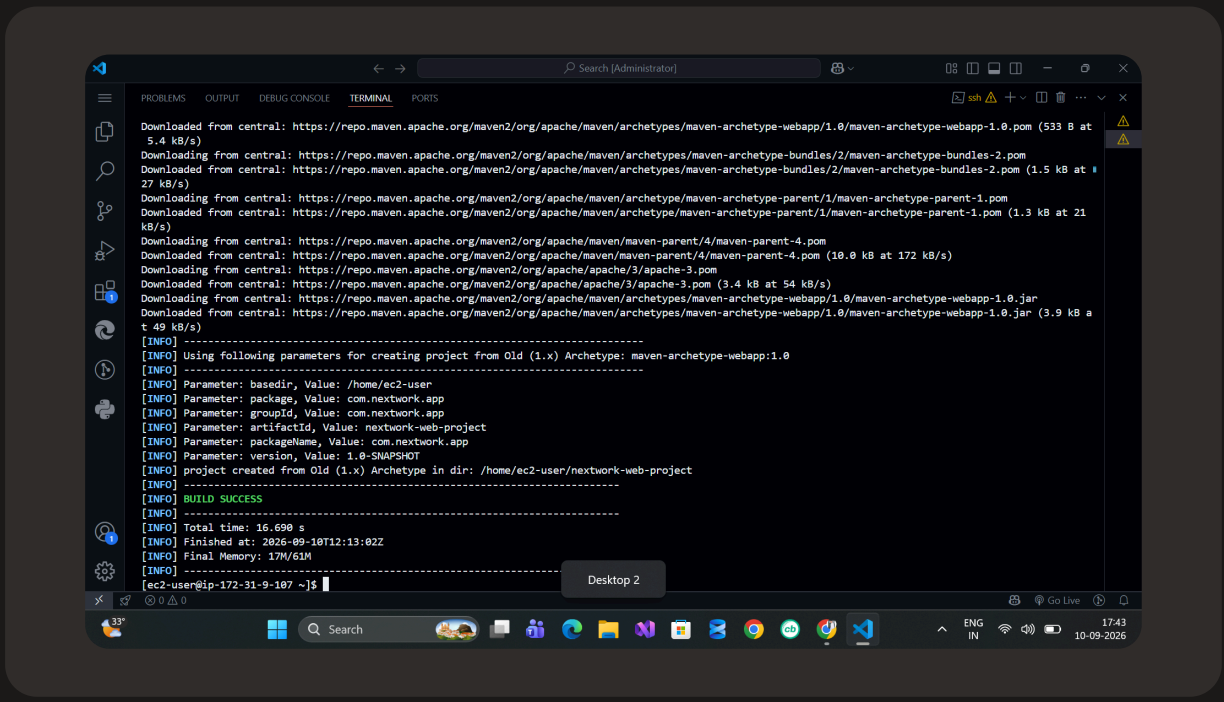
I generated a Java web app using the command listed above.

Installing Remote - SSH

I installed Remote - SSH, which is it lets you connect directly via SSH to another computer securely over the internet. I installed it to vs code using vs code extensions.

SSH configuration details

Configuration details required to set up a remote connection include HostName, IdentityFile and the User.

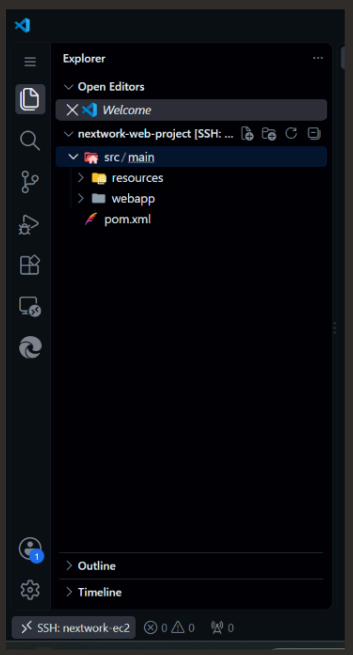


Create the Application

Exploring the project structure

Using VS Code's file explorer, I could see a folder in which there is a src folder which containes resource and webapp and a pom.xml file to handle and add the dependencies.

Two of the project folders created by Maven are src and webapp, which are the web app's files e.g. HTML, CSS, JavaScript, and JSP files, and resources, which are the configuration files a web app might need e.g. connection settings to a database.



Using Remote - SSH

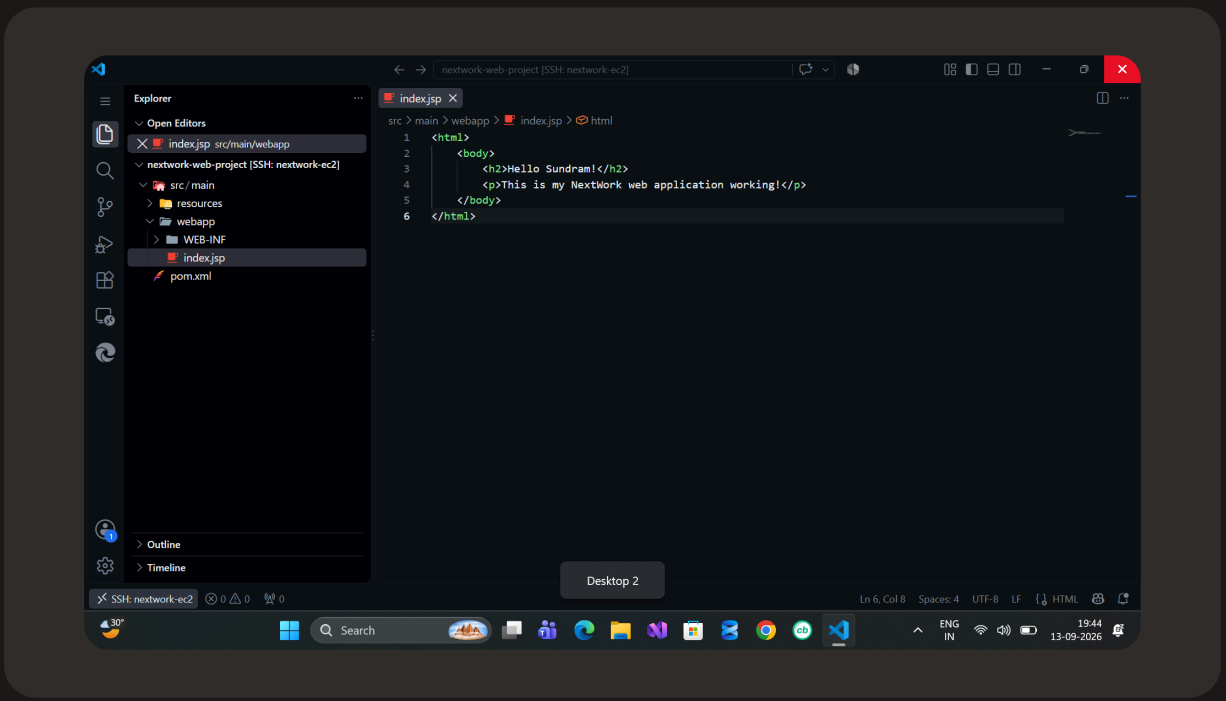
What I did in this step

In this step, I will download an extension of vs code to connect EC2 directly with the IDE.

Updating the web app

The index.jsp is a markup which we can use to make dynamic webpages and use java code inside that.

I edited index.jsp by adding h2 tag and paragraph tag and then save the updated file.



Using nano

Additional improvements

In this secret mission, I will change my .jsp file using the terminal

Verifying my work

To verify my editing work in the terminal, I added a line of code using terminal. It was possible to see my changes in VS Code right away because i added a line of code in .jsp file